THE REFLEXIVE HULL AS AN INTERSECTION OF LOCALISATIONS AT HEIGHT ONE PRIMES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The localisation of the module modulo its torsion at a height one prime is finite free over the discrete valuation ring, so tensoring the sequence relating torsion and duals is exact. Lemma \ref{lemma-dedekind-torsion-free-flat}, cited at proof lines 5656-5658. Identifies the terms appearing in the description.
- Criterion for reflexivity over a Noetherian normal domain via the intersection of the height one localisations. Lemma \ref{lemma-reflexive-over-normal}, cited at proof lines 5661-5662. Supplies the intersection statement that finishes the proof.

Source reference. Original lemma 0AVC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13680–13680. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0AUW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13664–13664. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0AVB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13679–13679. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

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Lemma 0.2. *Let R be a Noetherian normal domain. Let M be a finite R-module. Then the reflexive hull of M is the intersection*

$$M^{**} = \bigcap\nolimits_{height(\mathfrak{p})=1} M_\mathfrak{p}/(M_\mathfrak{p})_{tors} = \bigcap\nolimits_{height(\mathfrak{p})=1} (M/M_{tors})_\mathfrak{p}$$

taken in $M \otimes_R K$.

Proof. Let $\mathfrak{p}$ be a prime of height 1. The kernel of $M_\mathfrak{p} \to M \otimes_R K$ is the torsion submodule $(M_\mathfrak{p})_{tors}$ of $M_\mathfrak{p}$. Moreover, we have $(M/M_{tors})_\mathfrak{p} = M_\mathfrak{p}/(M_\mathfrak{p})_{tors}$ and this is a finite free module over the discrete valuation ring $R_\mathfrak{p}$ (Lemma 0AUW). Then $M_\mathfrak{p}/(M_\mathfrak{p})_{tors} \to (M_\mathfrak{p})^{**} = (M^{**})_\mathfrak{p}$ is an isomorphism, hence the lemma is a consequence of Lemma 0AVB. □